Vaccine Storage Optimization

Group 2

Rashedul Islam Sabbir

Nahin Mahmud Mahi

Monjurul Kadir Muid

Tawshif Alam Nahin

Ashikul Islam Ashik

Md Tahsin Rahman

MD Anjum Ashraf

Fahid ahmed

Introduction

Vaccines are biologically sensitive compounds. Their effectiveness varies on the temperature. So, the storage environment must be strictly maintained at a temperature where the vaccine's efficacy is at its absolute peak.

To analyze this, we have defined a vaccine efficiency model given by the function:

T(x) = $x^4 - 8x^2 + 10$

In this model, T(x) represents the efficiency of the vaccine, and X represents the storage temperature.

In the next section, we will demonstrate how we apply calculus to mathematically pinpoint the exact temperature X that yields the maximum possible efficiency.

Methodology

To solve the problem of optimization, we utilized Differential Calculus. Calculus allows us to analyze the behavior of functions and identify points where a system reaches its maximum or minimum values. By doing 1st Derivative of T(X) by using power rule we will get Critical points. The points where Extrema or Minima has occurred.

Figure 1: Value of T(X) in table

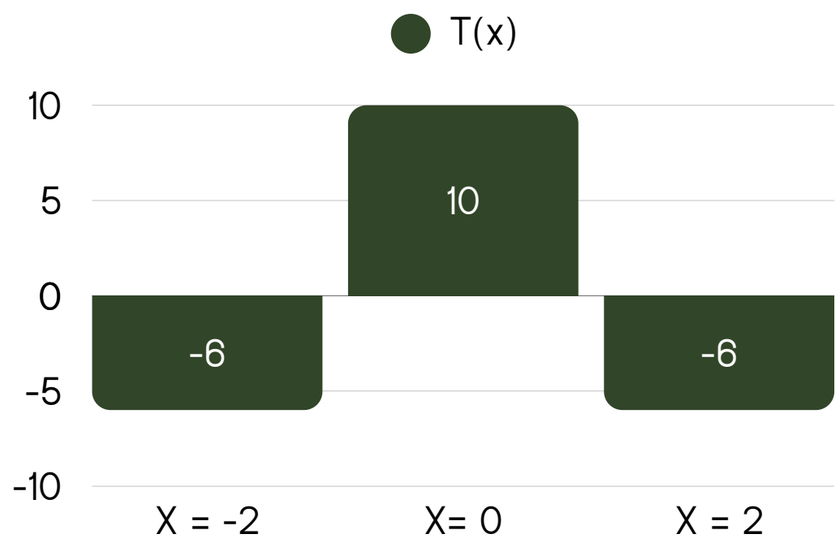
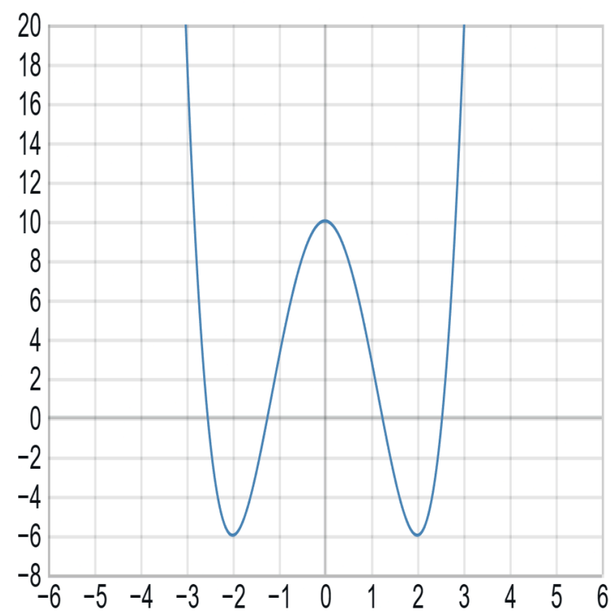


Figure 2: Graph on T(X)



Results

We have T(x) = x^4- $8x^2$ + 10 . By doing differentiation we will get T'(x) = $4x^3$- 16x . Then to find the critical point T'(x) = 0 . Then with algebra we will get x = -2 or 0 or 2

By plotting this critical point to our vaccine function T(x) we will get Efficiency at X temp on the vaccine:

T(-2) = -6

T(0) = 10

T(2) = -6

Conclusion

We successfully applied the principles of differential calculus to solve the real world problem of vaccine storage optimization. By differentiating the efficacy model T(x) = x^4- $8x^2$ + 10, we identified three critical temperature points: x = -2, x = 0, and x = 2.

Through mathematical verification, we determined that: x = -2 and x = 2 represent points of failure, where vaccine efficacy drops to a minimum value of -6 and x = 0 make the vaccine efficacy to maximum of 10

Therefore, we conclude that to maintain the highest standard of immunization effectiveness, the storage environment must be strictly regulated to target x = 0. This mathematical proof reinforces the importance of precision in medical logistics and demonstrates how abstract mathematical concepts, such as derivatives and extrema, are fundamental to preserving public health.

Abstract

Keeping vaccines at the right temperature is a huge challenge. If a vaccine gets slightly too hot or too cold, it breaks down and stops working. In this project, we used math to find the perfect storage temperature.

We used the formula T(x) = x^4- $8x^2$ + 10 to predict how well a vaccine works at different temperatures X. Using calculus, we calculated the derivative to find the exact peak of this curve. Our math proved that 0 degrees is the absolute best temperature. It also showed that if the temperature shifts to 2 or -2, the vaccine fails. This project shows how we can use calculus to solve real-world problems and keep medical supplies safe.

References

1. CDC Vaccine Storage Toolkit
   - Source: Centers for Disease Control and Prevention (CDC)
   - Why we used it: This guide explains that vaccines stop working if they get too hot or too cold. It proves that our problem is real and important.
2. Calculas: Calculas Anton 10th edition
   - Why we used it: This is a standard math book. We used it to learn the rules of Derivatives and how to find the Maximum point on a graph.
3. WHO Immunization Guide
   - Source: World Health Organization (WHO)
   - Why we used it: This manual shows that once a vaccine is ruined by heat, it cannot be fixed. This supports our conclusion that finding the perfect temperature is necessary.